



Recent advances in machine/deep learning frameworks for biochar application in soil amendment and remediation[☆]

Muhammad Waqas^c, Mohsin Nawaz^d, Rongli Yang^{a,b}, Waseem Tariq^{a,b},
Muhammad Nauman Tahir^{a,b}, Shakeel Ahmad^{a,b,*}

^a Yunnan Provincial Key Laboratory of Soil Carbon Sequestration and Pollution Control, Faculty of Environmental Science and Engineering, Kunming University of Science and Technology, Kunming, 650500, China

^b Yunnan International Joint Laboratory for Emission Reduction and Carbon Sequestration in Agricultural Soils, Kunming, 650500, China

^c The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand

^d Faculty of Arts and Sciences, The Agha Khan University, Karachi, 74800, Pakistan

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ABSTRACT

Global industrialization has substantially increased carbon emissions, creating an urgent need for sustainable environmental remediation and carbon sequestration strategies. Biochar (BC), a carbon-rich product derived from thermochemical conversion of biomass, exhibits variable physicochemical properties that critically influence its environmental performance. Despite growing interest in BC for soil amendment and pollutant removal, the systematic integration of advanced artificial intelligence (AI), including machine learning (ML) and deep learning (DL), to optimize BC production and application remains underexplored. This review systematically analyzes recent supervised, unsupervised, and hybrid ML/DL frameworks, including random forests, support vector machines, convolutional neural networks, Long Short-Term Memory models, and generative AI, to predict yields, physicochemical properties, and environmental applications of BC. Furthermore, applications of ML/DL frameworks are examined to identify patterns in BC performance in soil fertility enhancement and pollutant removal efficiency. In addition, life cycle assessment and cost-benefit analyses using AI frameworks are critically evaluated. Finally, this review identifies key challenges, including the scarcity of high-quality datasets, limited model interpretability, and high computational demands, among others, and proposes potential solutions through explainable AI frameworks and cross-disciplinary collaborations. The integration of ML/DL frameworks into BC research provides actionable insights for optimizing production processes, improving soil remediation strategies, and accelerating sustainable agricultural applications.

Acronyms

BC	Biochar
AI	Artificial intelligence
DL	Deep learning
SVMs	Support vector machines
GANs	Generative adversarial networks
SVD	Singular value decomposition
RNNs	Recurrent neural networks
WNNs	Wavelet neural networks
RFs	Random forests
GBDTs	Gradient boosting decision trees

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Q-Learning	A reinforcement learning algorithm
LCA	Life cycle assessment
SHAP	SHapley Additive exPlanations
AutoML	Automated machine learning
Grid-RFs	Grid optimization-based random forests
HTC	Hydrothermal carbonization
ML	Machine learning
ANNs	Artificial neural networks
k-NN	k-Nearest neighbors
PCA	Principal component analysis
CNNs	Convolutional neural networks

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* Corresponding author at: Yunnan Provincial Key Laboratory of Soil Carbon Sequestration and Pollution Control, Faculty of Environmental Science and Engineering, Kunming University of Science and Technology, Kunming, 650500, China.

E-mail address: shakeel@kust.edu.cn (S. Ahmad).

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FFNNs	Feedforward neural networks
XGBoost	Extreme gradient boosting
DTs	Decision trees
LSTM	Long short-term memory
LIME	Local interpretable model-agnostic explanations
CBA	Cost-benefit analysis
IoT	Internet of things
GHG	Greenhouse gas

1. Introduction

Biochar (BC) is a carbon-rich material obtained from thermal, chemical, and hydrothermal conversion of biomass residues and wastes in an O₂-limited environment (Pimenta-Ocampo et al., 2024; Zhang et al., 2024e). BC is prepared from different feedstocks, namely agricultural waste (Ahmad et al., 2020), wood residues (Zhu et al., 2005), food waste (Ahmad et al., 2021), and bone waste (Ahmad et al., 2023), via pyrolysis (Ahmad et al., 2023), torrefaction (Zhang et al., 2018), hydrothermal carbonization (HTC) (Cai et al., 2016), and gasification (Yao et al., 2018), among which pyrolysis and HTC are widely used techniques. BC is cost-effective and has potential for sustainable environmental applications due to its large surface area, better stability, and electrical conductivity (Patro et al., 2024; Rao et al., 2024). BC applications include soil amendments and fertilizing agents, adsorbents, electrochemical energy materials, digestion additives, catalysts, and CO₂ sequestration (Ahmad et al., 2019; Luo et al., 2023). Such flexible applications highlight absolute market potential of BC. Global BC market is expected to grow from US\$170.9 M in 2021 to US\$587.7 M in 2030 at a Compound Annual Growth Rate of >13% (Ukoba and Jen, 2022). Although such success has been achieved, BC production technology is still developing. Challenges such as feedstock quality fluctuations, supply chain complexities, and process optimization have stalled large-scale production of BC (Liao and Yao, 2021). Traditional process development for BC production generally involves trial-and-error experiments with numerous feedstock and process condition tests, which are tedious and time-consuming (Ahmad et al., 2020).

In the past few decades, machine learning (ML) and deep learning (DL), which are subfields of artificial intelligence (AI), have been widely used in BC-related studies. Fig. 1 depicts the classification of AI-based frameworks. A comprehensive description of mechanisms used in ML/DL frameworks in BC research is provided in Text S1 of the Supplementary Information. ML/DL frameworks are robust and influential in solving real-world problems and have attracted significant attention due to their broad-field applications (Waqas et al., 2023). These frameworks enable systems to learn from data, illustrations, and experiences without predefined rules, transforming the international economy, environment, and society (Society, 2017). For example, DL employs multilayer artificial neural networks (ANNs) to optimize large datasets, and many DL algorithms are convolutional neural networks (CNNs), Long Short-Term Memory (LSTM) models, autoencoders, and generative adversarial networks (GANs), with their successful application in multiple agriculture and farm applications (Ren et al., 2024; Waqas et al., 2025; W. Zhang et al., 2023).

Previously, supervised ML/DL frameworks have been used for BC production (Li et al., 2022), classification (Jia et al., 2023), and soil applications (Dangi et al., 2020). Several studies have applied support vector machines (SVMs), grid optimization-based random forests (Grid-RFs), decision trees (DTs), random forests (RFs), and ANNs to predict removal of heavy metals (HMs), organic compounds, and dyes using adsorbents (Costa et al., 2017; LeCun et al., 2015; Li et al., 2022; C. Wang et al., 2024). Similarly, for BC characteristics such as yield, carbon content, and surface area, SVMs, RFs, and ANNs have been employed to predict important BC characteristics relative to composition of feedstock and pyrolysis parameters (Li et al., 2022). The BC adsorption capacities of HMs and organic compounds under various physicochemical conditions have been predicted using extreme gradient boosting (XGBoost) (C. Wang et al., 2024). CNNs and recurrent neural networks (RNNs) are being investigated as applicable image classification methods for BC morphology and time-series prediction of BC-amended soil performance (W. Zhang et al., 2023). In addition, hybrid approaches combining wavelet transforms with LSTM/RNNs have demonstrated the capability to model the temporal dynamics of BC's impact on soil carbon sequestration and greenhouse gas (GHG) emissions (Luo et al., 2023; L. Wang

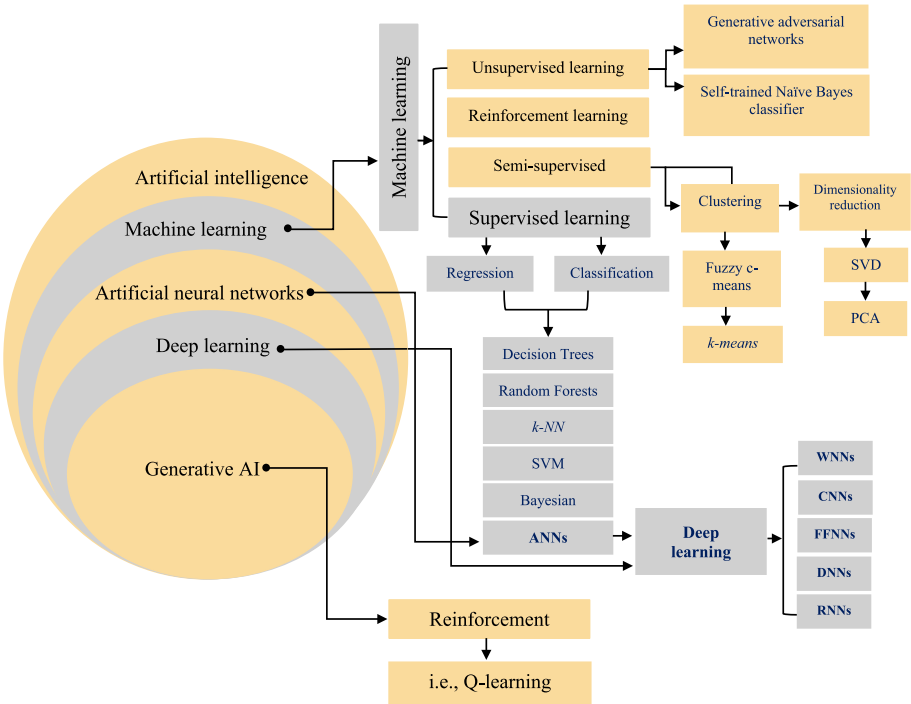


Fig. 1. Classification of AI-based methods, including ML/DL frameworks.

et al., 2023).

Abovementioned ML/DL frameworks are mainly used for BC preparation and application to remove HMs, dyes, and organic compounds from water and wastewater. However, their use for removal of HMs and organic compounds from soil is limited (Guo et al., 2023). Only a few studies have reported the nature of unsupervised learning frameworks, such as clustering and principal component analysis (PCA), for exploratory analysis and dimensionality reduction in BC datasets (Jia et al., 2023; Shafizadeh et al., 2023; M. Zhang et al., 2024). These approaches help classify BC types and functionalities by searching for hidden patterns in large datasets. Despite growing interest in BC applications, its integration into soil systems, particularly for soil amendment and remediation, has received limited attention within ML/DL frameworks. Existing reviews often lack a systematic assessment of how ML/DL can enhance predictive modeling and process control for BC performance and forecast soil interactions, considering the complex and nonlinear behaviors of BCs in soil. To address the need for a precise review in this field, the authors systematically compiled data on BC preparation, characterization, and soil applications, specifically soil amendment and remediation, and critically related these to available ML/DL frameworks.

This review aims to provide a thorough and systematic overview of ML/DL frameworks applied to BC preparation, characterization, optimization, and soil applications. It addresses the limitations of past studies by a comprehensive synthesis using PRISMA guidelines, including understudied DL frameworks, and highlights the importance of economic and interpretability aspects that were overlooked in previous reviews. Furthermore, Table S1 provides a detailed comparison of current review with previous reviews and highlights significance of this review. Specifically, this review provides a basic understanding of BC production via HTC and pyrolysis, its characterization with multiple techniques, and its application to soil amendment and remediation using ML/DL frameworks. It explores recent economic feasibility and environmental impact of BC application to soil. Finally, it outlines the prospects and challenges of integrating ML/DL into soil BC applications.

2. PRISMA methodology

PRISMA outline ensured a rigorous and reproducible review process (Sarkis-Onofre et al., 2021). A search strategy was designed to identify peer-reviewed articles on integrating ML and DL in BC production, characterization, and soil applications.

Boolean operators were used to refine search queries in two significant databases, Web of Science and SCOPUS, to source articles. Literature search was conducted using the Web of Science database with the following query:

("Machine Learning" OR "Deep Learning") AND ("Biochar preparation" OR "Pyrolysis" OR "Biochar production") AND ("Soil Amendment" OR "Soil Remediation").

The results were further refined by including MUST-HAVE keywords such as:

"Machine Learning", "Biochar", and either "Machine Learning Modeling" or "Deep Learning Modeling".

Additional keywords categorized as SHOULD-HAVE included:

"Biochar Soil Application," "Lifecycle Assessment," "Cost-Benefit Analysis," "Biochar Quantification," and "Biochar Characterization."

Initial search retrieved 575 records. A complementary search was conducted in the SCOPUS database using the following keywords:

("Deep Learning" OR "Machine Learning") AND ("Biochar Characterization" OR "Biochar Quantification") AND ("Soil Applications" OR "Lifecycle Assessment").

Only peer-reviewed articles published between 2015 and 2024 were included. Duplicate entries from both databases were programmatically removed using Python. Subsequently, irrelevant studies were excluded through manual screening of titles and abstracts. Further refinement was applied during the final review stage to eliminate studies that lacked substantial focus on either ML/DL methodologies or BC-related themes. After applying these criteria, 235 articles were retained in the final dataset. This systematic review followed the PRISMA framework, and the detailed methodology for integrating ML/DL frameworks in BC research is illustrated in Fig. 2.

3. BC preparation

This section explores the modification of biomass and preparation of BC waste via HTC and pyrolysis, using both traditional and microwave (MW)-induced heating methods. Implementation of ML/DL frameworks in BC research is an advanced approach to optimize production processes and improve environmental outcomes (Fig. 3). ML/DL frameworks combine multiple input parameters (preparation conditions (temperature, residence time, and moisture), material properties (carbon, nitrogen, and phosphorus content and ash to volatile ratios), and operational data (heating patterns and carbon metrics)) to generate predictive and prescriptive analytics (Sharma et al., 2023). Nonlinear relationships between process variables and BC characteristics are modeled using advanced ML/DL frameworks, such as ANNs and DNNs (Akinpelu et al., 2023; Al-Sakkari et al., 2024; Waqas et al., 2023). However, generative AI and optimization algorithms can simulate several pyrolysis scenarios, leading to the selection of optimal conditions for achieving maximum BC yield without substantial energy costs (W. Wang et al., 2024). In addition, ML frameworks contribute to BC characterization (quantity forecasting and quality control) by enabling predictive analysis of its physicochemical properties (e.g., surface area and pore structure) with best soil addition efficiency, carbon sequestration potential, and remediation performance (Nguyen et al., 2024b; Sharma et al., 2023).

3.1. Traditional and MW-induced HTC

HTC is a cost-effective method for modifying wet biomasses (e.g., sludge, manure, and agricultural residues). Unlike power-driven dewatering methods, HTC uses water as a reaction medium, avoiding additional operational costs while depolymerizing compounds and breaking chemical bonds through hydrolysis, dehydration, and condensation (Cai et al., 2016). This closed process prevents environmental hazards, making it environmentally friendly. HTC can be enhanced by traditional or MW-induced heating, each of which has distinct advantages and limitations. MW-induced HTC (MW-HTC) offers selective heating, reduces energy costs, and ensures uniform reactant distribution at mild temperatures and short reaction times (Ahmad et al., 2020). MW irradiation accelerates volumetric heating via molecular collisions, thereby improving processing efficiency and final product dewaterability. From an economic perspective, MW-HTC consumes less energy because of its rapid and thorough heating (Shao et al., 2019). HTC produces solid, liquid, and gaseous products that are rich in organics and nutrients. In HTC, reaction temperature typically ranges from 150 to 350 °C, with reaction times varying from minutes to hours. In MW-HTC, temperature ranges from 120 to 240 °C, and reaction time is shorter, ranging from seconds to minutes. Solid product, BC, contains carbonaceous structures with polar O₂ functionalities for better reactivity, enhancing surface functionality and calorific value. These characteristics make HTC BC suitable for use as an adsorbent (Ahmad et al., 2019), biofuel (Ahmad et al., 2021), soil amendment agent (Ahmad

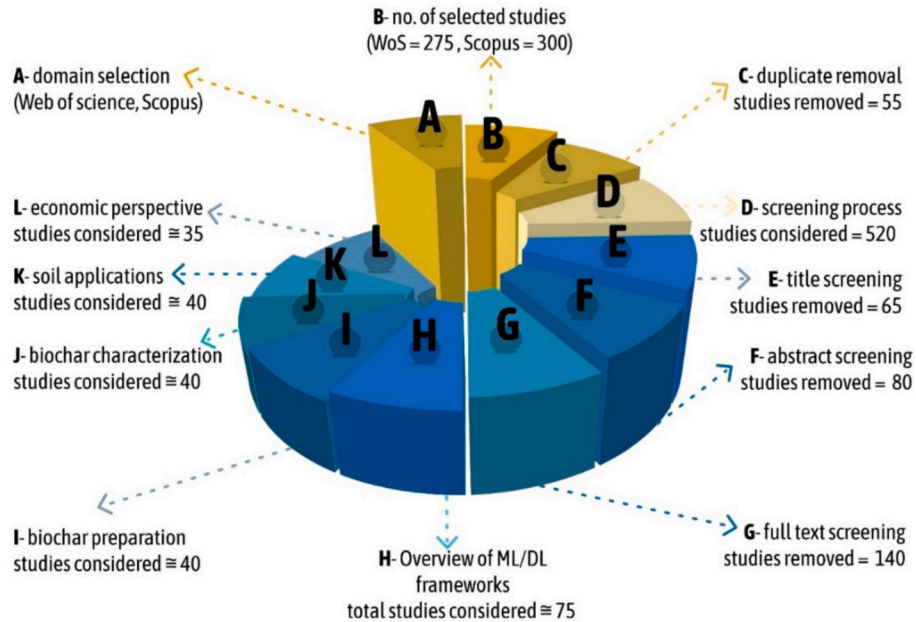


Fig. 2. PRISMA methodology and screening process of selected literature for integration of ML/DL frameworks in BC research.

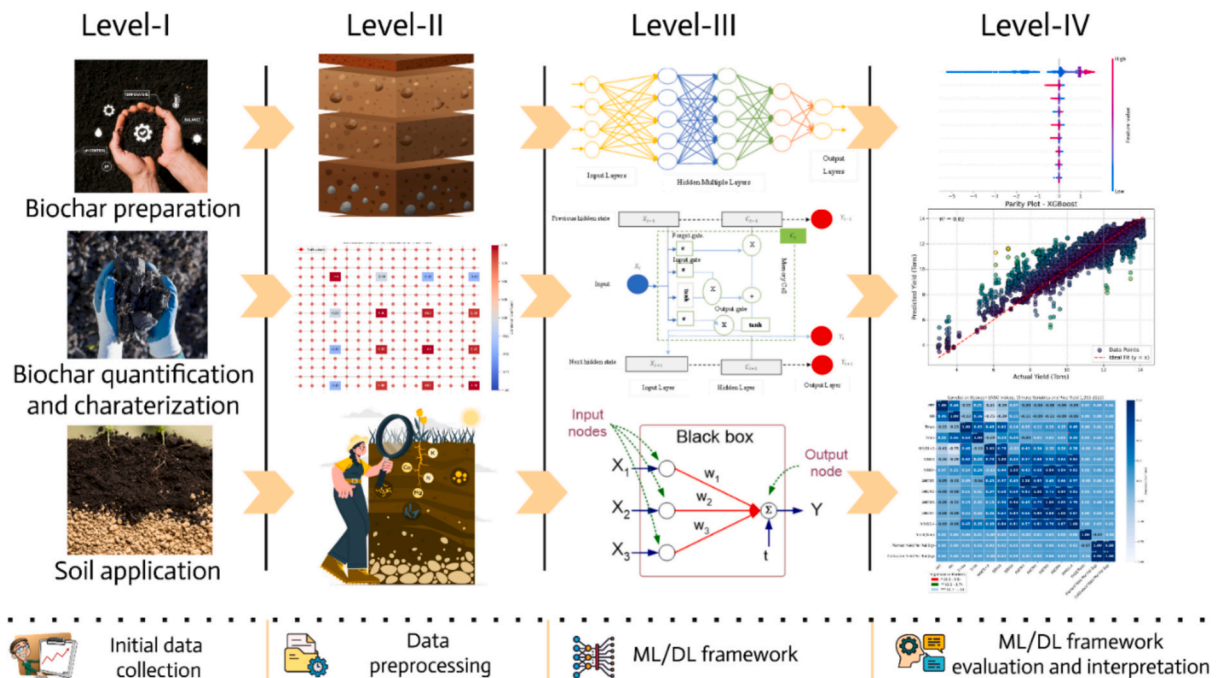


Fig. 3. Implementation of ML/DL framework in BC research.

et al., 2020), and carbon-based catalyst (Kang et al., 2013).

ML frameworks, such as RFs, SVMs, XGBoost, and ANNs, have been used to predict the critical properties and environmental applications of hydrothermal BC (Table 1) (Liu et al., 2024; Moussaoui et al., 2024). Best RFs achieved an average accuracy of 94.89%, and sensitivity analysis identified reaction temperature, pressure, and feedstock elemental composition as the most important parameters (C. Chen et al., 2023). Yuan et al. employed five DTs to predict the results of HTC and pyrolysis. RFs and XGBoost demonstrated high predictive accuracy (R^2 : 0.84–0.98) for various output variables, including nitrogen recovery, energy density, and residual sulfur, highlighting relevance of feedstock nitrogen and sulfur contents in determining BC quality and partial

dependence plots were employed to elucidate the relationships (Yuan et al., 2024). Another study integrated ML frameworks (ANNs, SVMs, and XGBoost) with response surface methodology to optimize HTC of almond shells and olive pomace. ANNs outperformed other models in predicting mass yield and carbon retention, and BC produced possessed promising adsorption abilities for methylene blue under optimized conditions (Moussaoui et al., 2024).

3.2. Traditional and MW-induced pyrolysis

Pyrolysis, conducted in a low- or no- O_2 atmosphere at 300–1000 °C, produces BC, oil, and gas (Ahmad et al., 2023). Higher temperatures

Table 1

ML/DL frameworks have been applied to predict BC production and properties for environmental applications.

ML/DL framework	ML/DL brief	Feedstock	Input variables	Output/application	Sources
XGBoost	Predicting BC yield and carbon content	Lignocellulosic biomass	Biomass properties, pyrolysis conditions, and carbon content	According to feature examinations and SHAP analyses, temperature was vital for BC yield and oxygen and carbon content	(Xu et al., 2024)
RFs	Predicting BC yield and carbon content	Lignocellulosic biomass	Pyrolysis conditions and biomass characteristics	Optimized BC production and understood biomass pyrolysis	(Zhu et al., 2019)
RFs	Regression	Kraft lignin	Carbonization characteristics (carbon content, O/C ratio, and H/C ratio)	Selective application of spectral regions was a better strategy for characteristic prediction	(Hwang et al., 2023)
RFs	Prediction of BC adsorption capacity	Various BC types	BC surface area, total carbon content, pH, and adsorption conditions	TC adsorption efficiency and kinetics	(P. Zhang et al., 2023)
SVMs-ANNs	Optimization of BC adsorption properties	Various BCs	BC characteristics, HMs concentration/sources, and environmental conditions	Relationships for BC adsorbents for enhanced HMs adsorption capacity	(Ke et al., 2021)
GBR	Estimating BC's long-term stability and removal rates	Various BCs	BC pH, CEC, elemental composition, yield, and surface area	Prediction feature engineering of BC for HMs adsorption	(Shen et al., 2024)
eXplainable ML models	Model interpretation and prediction	Biomass	Feedstock type and preparation temperature	Highlights advancements in sensitivity analysis, hyperparameter optimization, and ensemble techniques for BC production	(Nguyen et al., 2024a)
CNNs	Regression	Nitrogen-doped porous BC	Surface area, pore structure, and nitrogen content	Specific capacitance prediction for high-performance BCs	(Xiaorui et al., 2024)
GBT, RFs, SVMs, DTs, and MLR	Predicting N fixation rate and BC yield; GBT showed the best accuracy	Sewage sludge	Ultimate and proximate composition of sewage sludge	Prediction of nitrogen fixation rate (R^2 : 0.860) and BC yield (R^2 : 0.864); feature importance analysis (0.625 for N fixation rate)	(Li et al., 2023)
XGBoost, RFs, and SVMs	Predicting HTC BC fuel properties; XGBoost showed best performance (R^2 : 0.927, RMSE: 3.279)	Wheat straw	Ash content, reaction temperature, and solid-to-liquid ratio	BC fuel properties (R^2 : 0.825–0.985, RMSE: 1.119–5.426); optimization of HTC conditions	(Liu et al., 2024)
GBDTs	Tree-based models to predict SSA, TPV, MP, and NC accurately	Agricultural waste for nitrogen-doped porous carbon	Model-prediction of nitrogen content and other properties	GBDTs provided the best results with R^2 : 0.939	(Li et al., 2024b)
RFs, SVMs, and ANNs	Interpretable ML models enhanced with data preprocessing and sensitivity analysis; RFs with highest accuracy (94.89%)	Various biomasses	Reaction temperature and pressure, specific elements of biomass	Prediction of hydrothermal BC characteristics	(C. Chen et al., 2023)
BRT-SHAP (Boosted Regression Tree with SHAP)	Hybrid ML technique combining BRT, AdaBoost, and SHAP for improved interpretability and model transparency	Various biomasses	Multiple variables, including temperature (identified as most important by SHAP)	Prediction of BC characteristics (e.g., yield, proximate analysis)	(Le et al., 2023)
RFs	Supervised ML regression models	46 types of agricultural biomass	14 initial input variables, reduced to 9 using feature importance (e.g., pyrolysis temperature, etc.)	Prediction of BC yield and specific surface area, reducing the need for experimental work	(Hai et al., 2023)
ELT-PSO (Ensembled Learning Tree with PSO)	ML model optimized with PSO; compared with GA-integrated ML models	Biomass	Feedstock properties and operating conditions	Prediction and evaluation of BC yield; interaction analysis using PDPs; user-friendly software for yield estimation	(Haq et al., 2022)

break biomass into hydrocarbons, with phase transitions between 400 and 700 °C and aromatics condensing at 1000 °C. Temperatures >400 °C enhance BC properties, including graphitized layers, surface area, and pore structures (Ahmad et al., 2023). MW-induced pyrolysis employs MW radiation for volumetric heating, which is a rapid, non-contact, and energy-efficient process (Jeyasubramanian et al., 2021). It enhances safety and lowers operational costs compared to traditional pyrolysis. MW-induced pyrolysis is a less polluting and more efficient method for waste minimization and selective heating. Pyrolysis types, slow, fast, and flash, are distinguished by differences in heating rate, temperature, and reaction time (Ethaib et al., 2020). During pyrolysis, several gases, such as air, CO₂, and N₂, are supplied for proper operation (Liu et al., 2018).

ML/DL frameworks have enhanced process optimization of BC production in pyrolysis and other thermal techniques (Li et al., 2023; Zhu et al., 2019) (Table 1), with notable implications for soil remediation. For example, pyrolysis conditions and BC properties have been optimized using ML, and the soil profile was improved after BC application (Akinpelu et al., 2023). Similarly, ANNs represent nonlinear

dependencies among feedstock composition, pyrolysis conditions, BC properties, and adsorption capacity, optimizing pyrolysis to enhance BC quality and predict adsorption kinetics of HMs and organic compounds (M. Zhang et al., 2024). RFs and SVMs have also mapped feedstock properties to pyrolysis conditions to predict BC yield, surface area, adsorption capacity, and soil fertility enhancement; combined with generative AI (e.g., GANs), these approaches can forecast BC's efficacy in removing HMs and organic contaminants from soil and water using large ML datasets (Lakshmi et al., 2021). Without extensive laboratory work, GANs can generate synthetic datasets to overcome data scarcity and explore new feedstocks or pyrolysis conditions (Ren et al., 2024). SVMs have been applied to predict the yields of pyro-products in co-pyrolysis of biomass and plastics and identify significant physicochemical properties (carbon, ash, and volatile matter) that can be extrapolated to soil matrices (Potnuri et al., 2023). Moreover, SHAP analysis revealed that temperature and MW power were dominant features that impact yield outcomes, similar to those in contaminant degradation in soil remediation (Yang et al., 2022). Large language models coupled with ANNs to optimize bio-oil yield from MW-induced pyrolysis offered a novel

approach that integrates language-based and data-driven insights to improve predictive accuracy and interpretability (B. Chen et al., 2024). Gradient Boosting Regression (GBR) models have proven to be highly accurate ($R^2 > 0.91$) in modeling the effect of thermal parameters of MW-induced pyrolysis of lignocellulose biomass (Sinha et al., 2024).

Pyrolysis conditions showed a greater influence than biomass characteristics on BC yield and carbon content. RFs outperformed multilayer perceptron (MLP) models and predicted the carbonization characteristics (carbon content, O/C ratio, and H/C ratio) of BC from kraft lignin HTC, achieving high accuracy with R^2 of 0.989, 0.988, and 0.985, respectively (Hwang et al., 2023). Models trained on high-importance spectral regions outperformed those using the full spectrum, highlighting the value of selective data application. RFs demonstrated highest accuracy (R^2 : 0.962) for optimal tetracycline (TC) adsorption onto BC with a specific surface area of $> 380 \text{ cm}^2 \text{ g}^{-1}$ and particle size of 2.5–14.0 nm (P. Zhang et al., 2023). TC-to-BC ratio was the most influential factor (weight: 0.595), with the concentration gradient as the primary adsorption driver. Twenty hybrid ensemble models, combining five ML algorithms (SVMs, RFs, ANNs, M5Tree, and GP) with bagging technique, were developed to predict HMs adsorption efficiency onto various BCs (Table 1) (Ke et al., 2021). These ensemble models outperformed individual models, demonstrating high reliability and the promise of hybrid AI for optimizing BC-based HMs removal.

4. BC quantification and characterization

Chemical and physical characterization of BC, using various techniques to assess its physicochemical, structural, and nutrient properties, is critical for understanding its performance in water treatment, pollutant removal, and soil and agricultural applications (Elkhilifi et al., 2023). These characterizations further aid in elucidating conversion of biomass in HTC and pyrolysis. Precision, scalability, and automation of BC physicochemical property identification have been achieved by adding ML/DL frameworks to quantification and characterization of BC for sustainable agricultural practices and carbon sequestration (Chen et al., 2023). For example, eXplainable ML models predict environmental impacts of feedstock type and preparation temperature on BC, such as its effects on soil health, microbial communities, and potential for carbon sequestration (Nguyen et al., 2024a). AI-based decision support systems provide optimal application rates, feedstocks, and preparation conditions considering local soil and climate conditions (Kakhani et al., 2023; Palansooriya et al., 2022). AI-based forecast models have indicated that BCs may alter soil health, including pH, organic matter content, and microbial activity (Palansooriya et al., 2022).

4.1. Proximate and ultimate analysis of BC

Proximate and ultimate analyses are basic compositional data analysis techniques, whereas spectroscopic and imaging techniques are structural and morphological attribute techniques. Elemental compositions (carbon, hydrogen, oxygen, and nitrogen) are determined because these elemental profiles correlate with BC performance metrics, enabling ML frameworks to identify optimal applications for soil amendment and contaminant removal (Cong et al., 2023; Zhou et al., 2022). RFs (Zhu et al., 2019) and XGBoost (Xu et al., 2024) algorithms predicted BC yield and carbon content in lignocellulosic biomass pyrolysis, providing a computationally efficient approach to optimize BC yield (Table 1). Researchers have employed ML frameworks (ANNs, DTR, support vector regression (SVR), Gaussian process regression (GPR), RFs, random forest regression (RFR), Multi-Linear, and Polynomial Regression) to estimate higher heating value of fuels from largest proximate and ultimate analysis datasets, covering biomass, BC, waste, oils, and gases on a dry and ash-free basis (Xing et al., 2019; Yaka et al., 2022). Along with ML frameworks, statistical values of ultimate analysis for each fuel class were examined, and an extensive Krevlen diagram

was created to illustrate relationships between atomic O/C and H/C ratios of BCs (Yaka et al., 2022). Furthermore, in ultimate analysis, carbon and hydrogen fractions were the most significant, with carbon having a significant influence, whereas oxygen and nitrogen showed limited effects (Xing et al., 2019).

4.2. Pore structure and surface area of BC

Porosity and surface area of BC depend on its performance as a soil amendment or pollutant adsorbent. Pore structure is commonly measured using Brunauer-Emmett-Teller (BET) analysis; however, combining CNNs with scanning electron microscopy (SEM) images enables quick and accurate quantification. Raj et al. applied DL frameworks to precisely segment BC micrographs to distinguish macro-, meso-, and micropores (Raj et al., 2021). These new developments help improve understanding of BC adsorption capacity and its interactions with soil microbiota. DL frameworks increase accuracy by automating feature extraction from spectral patterns and reducing reliance on manual interpretation. Simultaneously, DL-based image analysis has revolutionized physicochemical characteristics (pore structure/surface morphology) of BCs. High-resolution imaging techniques, including microscopy and spatial transcriptomics, generate multi-dimensional datasets that CNNs analyze to quantify heterogeneity of pores and surface roughness (Liu et al., 2019). These models capture structural changes linked to pyrolysis conditions, aiding rapid quality control and optimization of BC for soil use. However, standardizing imaging protocols and modeling the diverse nature of BC feedstocks remain challenging. Considering pore structure and surface area of BCs, extreme temperature and preparation conditions correlate highly with these BC properties. Alkalinity of BC increases with an increase in pyrolysis temperature because of alkaline earth metals in biomass feedstocks (Jegan et al., 2020).

4.3. Morphology of BC

SEM and transmission electron microscopy (TEM) image-based characterization techniques provide information regarding the physical structure of BC, such as porosity and surface morphology (Zou et al., 2020). CNNs have reduced BC microstructure assessment from SEM and TEM images to visual inspection (Nguyen et al., 2022). DL was used to analyze the texture analysis to describe its surface roughness and morphological heterogeneity. CNNs have been trained to classify BC structures under various conditions, standardize quality assessment, and optimize pyrolysis parameters using high-resolution SEM (Palansooriya et al., 2022). Similarly, image analysis and classification tasks have been applied to BC research using DL, particularly CNNs. For example, Jia et al. used CNNs to classify BC based on physical and chemical properties obtained from microscopic images (Jia et al., 2023). This has made BC studies more reproducible by eliminating observer bias and enabling objective quantitative assessments. Furthermore, DL is used to optimize BC for HMs adsorption or water retention in agricultural soils for BC image analysis applications (Kumari et al., 2024). Scientists using big data and sophisticated image processing algorithms have transformed BC production to achieve the required physical properties, making BC suitable for soil and environmental purposes (Lakshmi et al., 2021).

4.4. Spectroscopic analysis of BC

Spectroscopic techniques are the primary tools for providing information on the chemical structure, functional groups, and compositional attributes of BCs (Jia et al., 2023; Raj et al., 2021). ML has been applied to analyze and interpret spectroscopic data to create automated techniques for detecting and classifying BC properties (Burud et al., 2016). Fourier transform infrared (FT-IR) spectroscopy is used to analyze functional groups in BCs, providing insights into their chemical composition and stability (Ahmad et al., 2019). Interpreting FT-IR data

is time-consuming and requires considerable expertise. Incorporation of ML frameworks also increases reliability of FT-IR-based BC classification, which can distinguish BC from different feedstocks and preparation conditions (Amin et al., 2016). Hi-tech spectroscopic methods, such as UV–vis absorption and solid-state ^{13}C nuclear magnetic resonance (NMR), generate massive amounts of data that ML algorithms can decipher to predict dissolved organic carbon concentrations and aromaticity. For example, UV–vis spectral slopes and absorption can be used by ML frameworks to classify BC-dissolved organic carbon fractions without extensive wet chemistry (Liu et al., 2019). BC has also been profiled using NMR spectroscopy to establish its aromaticity and structural composition. ML algorithms, such as RFs and GBT, have identified correlations between NMR spectral features of BC and its stability and carbon sequestration potential (Brewer et al., 2011).

4.5. Nutrient analysis of BC

ML/DL frameworks are used to establish relationships between BC properties and nutrients. Adding sewage sludge BC to soil accurately predicted water-soluble organic matter and its associated nutrients, restoring damaged soils [96]. Chen et al. recommended the use of FT-NIR spectroscopy and back propagation neural DL to estimate soil nutrient content. Competitive adaptive reweighted sampling reduces the number of spectral bands, thereby improving the signal-to-noise ratio. Because back propagation neural DL can identify abstract feature structures, it is more accurate and reliable than principal component regression and partial least squares regression (PLSR) (Chen et al., 2018). After adding BC, CNNs have been used to measure changes in nitrogen, phosphorus, and potassium (Jin et al., 2023). Nitrogen and phosphorus are essential nutrients that enhance soil fertility and agricultural productivity through BC applications. Basic techniques, such as elemental analysis, energy-dispersive spectroscopy, and element mapping, have been employed to estimate nitrogen in BCs. Furthermore, FT-IR and X-ray photoelectron spectroscopy have been used to analyze nitrogen functional groups in BCs (Ahmad et al., 2020). Phosphorus and its functional groups in BCs have been investigated using a range of established methods, including Hedley's sequential fractionation method, Standards, Measurements, and Testing protocol, X-ray absorption near edge structure spectroscopy, and ^{31}P NMR analysis (Ahmad et al., 2022). RFs, SVMs, and XGBoost predict BC nutrient

content from pyrolysis, feedstock, and reactor information, identifying temperature, residence time, and ash content as key predictive factors (P. Zhang et al., 2024). Using ML/DL with reinforcement learning, AI has been applied to enhance nutrient supply in anaerobic digestion with BC (Rajpurohit et al., 2025). Despite progress, findings from specific BC-soil systems remain hard to generalize for different types. Integrating chemical reactions with multimodal data (spectral, imaging, and chemical) could enable knowledge transfer and improve nutrient prediction across diverse systems.

5. Soil applications of BC

Fig. 4 shows the overview of AI frameworks used in literature for different BC applications. It is widely accepted that BC improves soil fertility, water retention, and carbon sequestration while reducing GHG emissions (Ahmad et al., 2024). It also immobilizes HMs and organic compounds, acts as fertilizing agent, and lowers environmental risks in soil, making it valuable for sustainable soil management, amendment, and remediation with the assistance of ML/DL frameworks (Table 2).

5.1. Soil amendment

Using biomass and BC as fertilizers and soil amendments is crucial for sustainable agricultural practices (Ahmad et al., 2022). BC increases soil structure, pH, water retention, cation exchange capacity (CEC), fertility, and nutrient uptake, and decreases GHG emissions (Ahmad et al., 2024). It also affects microbial composition, nutrient cycling, and supply. BCs contain high amounts of macronutrients (Ca, K, Mg, Na, and P) and micronutrients (Fe, Mn, and Cu), which are important for crop growth (Dangi et al., 2020; Qian et al., 2016). Phosphorus is significant since 80% of phosphates are applied to fertilizers (Manolikaki et al., 2016). Mixing BC into soil for rice cultivation reduced nitrogen and phosphorus fertilizer use by 30% and 50%, respectively (Mohammadi et al., 2016). Fertilization with BC reduced N-P-K fertilizer applications by 50% and improved sunflower yield (Furtado et al., 2016). HTC and pyrolysis increase the molecular activation and dewaterability of biomass waste to support plant growth. Sewage sludge-derived BCs enhanced ryegrass germination (Paneque et al., 2019), and BC increased corn's nitrogen and potassium uptake in sandy soils (Syuhada et al., 2016). Compared to control, BC-treated field soil increased plant height

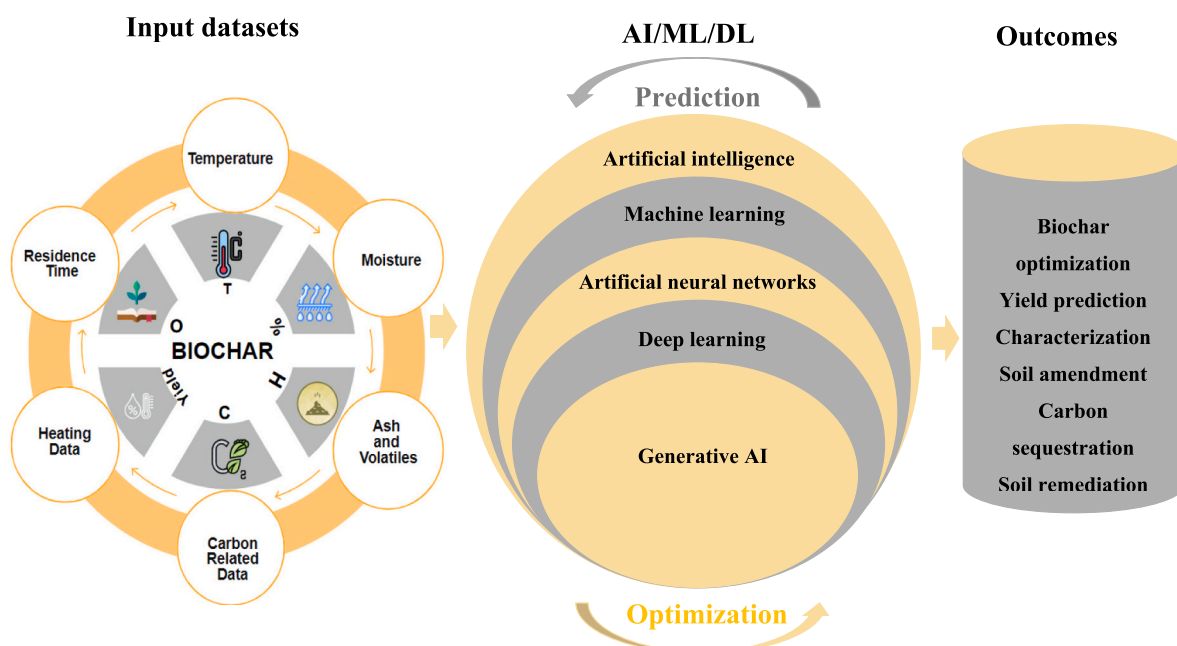


Fig. 4. Overview of AI (ML/DL) frameworks used in literature for different applications in BC research.

Table 2

ML/DL frameworks applied to predict BC's effectiveness in soil amendment and soil remediation.

ML/DL framework	ML/DL brief	Feedstock	Input variables	Output/conclusion	Sources
DTs and ensemble learning	Prediction of soil carbon sequestration potential with BC application	Various BC feedstocks	SOC, BC stability, and decomposition rate	Long-term carbon sequestration estimation in soils	(Tsolis and Barouchas, 2023)
Five ML frameworks	MLP-NN, DTs, SVMs, ensembles of trees, and GPR	Various BC feedstocks	Feedstock proximate and ultimate composition and pyrolysis reactor conditions	Better model performances; BC yield and carbon/nitrogen content (R^2 : 0.92–0.97)	(Li et al., 2025)
XGBoost and LGB	Regression	–	Soil moisture, electrical conductivity, soil temperature, and weather parameters	Soil moisture prediction and irrigation/water management.	(Teshome et al., 2024)
GBDTs	Regression	–	Soil pH, CEC, clay content, and SOC	HMs (Cd, Cr, Cu, Pb, Ni, and Zn) adsorption capacity prediction	(Yang et al., 2021)
GBR	Regression	Various BCs	BC pH, CEC, elemental composition, yield, and surface area	Prediction feature engineering (long-term stability and removal rates) of BC for HMs adsorption	(Shen et al., 2024)
LSTM, BiGRU, 5-layer CNNs	LSTM: sequence-based time series model; BiGRU; Deep CNNs for spatial feature learning	BC	Soil pH, clay content, CEC, SOC, and electrical conductivity	In control soils, CNNs consistently showed the highest precision in predicting Cd adsorption risk.	(Rashid et al., 2024)
SVR	Regression	–	Six factors: DOC, BC content, SAC, GWC, DS, and porosity Six factors: degree of compaction, BC content, soil air content, gravimetric water content, degree of saturation, and porosity	Gas diffusion coefficient prediction in BC-amended soils	(Onyekwena et al., 2022)
ANNs	Predicting phytoremediation efficiency	Municipal waste compost-BC blend	Soil physicochemical properties (pH, EC, CEC, and SAR) and BC properties	HMs immobilization in contaminated soils (highest concentrations at 0.8% compost (274.82 mg kg ⁻¹ Zn) and 0.52% BC (26.66 mg kg ⁻¹ Cr))	(Roohi et al., 2020)
Hybrid ML/DL models	Optimizing of CNNs-LSTM and LSTM-CNNs with BOHB	Not applicable (Vis-NIR spectra of forest soil)	In situ Vis-NIR spectra, organic matter content, and augmented spectral datasets	BOHB-optimized hybrids outperformed traditional models (PLSR and RFs)	(X. Wang et al., 2023)
DNDC model	CH ₄ emission prediction in paddy soils with BC and nitrogen addition	Rice husk BC	Nitrogen fertilization rates, BC percentage, soil microbial activity, and soil pH	Reduction in CH ₄ emissions from paddy soils	(Shaukat et al., 2022)
AutoML	H ₂ O AutoML for predicting eight remediation variables	–	Remediation technology, organic carbon, organic nitrogen, phosphates, petroleum hydrocarbon concentration and type, moisture, and cultivation period	Remediation efficiency of soil enhanced with petroleum hydrocarbon concentration < 5000 mg kg ⁻¹	(J. Wang et al., 2024)
ML-based soil monitoring	Predicting BC impact on soil microbial diversity and HMs immobilization	BC from mixed wood biomass	Microbial community composition, enzyme activity, BC's nitrogen content, application rate, soil pH, organic matter content, and HMs properties	Soil fertility enhancement and HMs immobilization efficiency in BC-amended soils	(Palansooriya et al., 2022)

and biomass by 20% and 37% respectively (Baronti et al., 2017). Manure-derived BC application at 15–20 t ha⁻¹ in soil increased corn yield by 150% (Uzoma et al., 2011), and poultry litter-derived BC application at 10 t ha⁻¹ increased radish yield by 42% (Chan et al., 2007). BC also alters soil microbial communities, and its shape and pore size maintain stability for microbial activities, supporting long-term soil health (Thies and Rillig, 2009). BC reduced *acidobacteria* >5 folds and increased *proteobacteria* >1.6 folds, shifting bacterial and fungal communities and affecting soil organic carbon (SOC) decomposition (Andert and Mumme, 2015). Sewage sludge-derived BC improved *phosphomonoesterase* activity by 70.8%, promoting nutrient mineralization and root development (Paz-Ferreiro et al., 2012). Chicken manure-derived BC enhanced *dehydrogenase* activity, increasing microbial activity from 2.75 to 8.96 mg TPF kg⁻¹ 24 h⁻¹, indicating that BC influences microbial communities (Park et al., 2011).

Studies have employed DL frameworks, DNNs, CNNs-RFs, and VGG-RNNs to assess soil quality, nutrients, pH, and microbial interactions to improve soil amendment practices (Escorcia-Gutierrez et al., 2022; Pawar et al., 2022; Shen et al., 2024). Proposed ISnPHC-WVE model outperformed baseline frameworks, achieving 92.81% and 94.97% accuracy for nutrient and pH classification, respectively (Escorcia-Gutierrez et al., 2022). Moreover, Pawar et al. confirmed that CNNs-RFs and VGG-RNNs showed better classification ability with accuracies of 95.8% and 97.9%, respectively, demonstrating the potential of hybrid federated architectures in soil quality monitoring (Pawar et al., 2022). DNNs are suitable for evaluating BC yield, HMs immobilization, carbon

sequestration, and soil microbial interactions (Ke et al., 2021; Shen et al., 2024; M. Zhang et al., 2024). ML framework integration has helped increase BC classification precision, adsorption efficiency estimations, and long-term stability assessments (Yang et al., 2021; P. Zhang et al., 2023). BC and municipal waste compost, along with *bromus tomentellus*, were evaluated for phytoremediation of Cr⁶⁺ and Zn²⁺ contaminated soils, with application of ANNs to predict HMs extraction (R^2 : >0.9) (Roohi et al., 2020). ANNs successfully extended the range of amendment predictions from 2% (experimental) to 5%, demonstrating their utility for optimizing soil amendment strategies. ML frameworks were tested to predict the gas diffusion coefficient of BC-amended soil under varying conditions, accounting for six factors: degree of compaction, BC content, soil air content, gravimetric water content, degree of saturation, and porosity (Onyekwena et al., 2022). Epsilon radial basis function support vector regression model performed best (R^2 : 0.9925). Soil air content, degree of saturation, and porosity were identified as main predictors, with soil air content being the most influential. An optimal gas diffusion coefficient was established for different degrees of compaction and BC levels, aiding in the design of engineered landfill covers. To predict soil carbon sequestration potential with BC application, potential of VIS-NIR spectroscopy with DTs and ensemble learning for monitoring BC-soil interactions highlights complexity of SOC, BC stability, and decomposition rate relationships, which vary with environmental conditions for long-term carbon sequestration estimation in soils (Tsolis and Barouchas, 2023).

Among five ML frameworks, the MLP and GPR best predicted BC

yield and carbon/nitrogen content (R^2 : 0.92–0.97), with the former identifying optimal pyrolysis conditions (400 °C, 15 °C min⁻¹, and 40 min) for maximum GHG savings (Li et al., 2025). These robust ML frameworks demonstrated the ability to optimize pyrolysis conditions to maximize BC's soil applications. ANNs and RFs predicted metal immobilization via BC amendment, with BC properties (34.3–53.1%) outweighing soil properties, and BC dose and pH being the most influential factors, enabling evaluation of 75 real-soil amendment outcomes (Sun et al., 2022). Few studies have developed DL/ML frameworks, such as XGBoost, LGB, RFs, k-nearest neighbors (k-NN), and LSTM, for sub-hourly soil moisture and HMs adsorption prediction. ML frameworks, particularly XGBoost and LGB, outperformed DL and LSTM, achieving higher R^2 (0.86–0.88) and better accuracy for data from two years on green beans and sweet corn under varying irrigation treatments (Teshome et al., 2024). In addition, hybrid ML models and process-based simulations (e.g., denitrification and decomposition (DNDC) model) have been applied to predict CH₄ emissions and BC decomposition rate in soil environment (Shaukat et al., 2022) (Table 2). Hybrid DL frameworks (CNNs-LSTM and LSTM-CNNs) adjusted with Bayesian optimization with HyperBand (BOHB) outperformed traditional methods in predicting soil organic matter from Vis-NIR spectra of 600 forest soil samples, showing strong promise for soil property prediction from large spectral datasets (X. Wang et al., 2023).

These studies demonstrate that ML/DL frameworks offer scalable, high-accuracy solutions for optimizing soil amendment and nutrient strategies under complex and heterogeneous conditions by integrating multidimensional soil, environmental, and physicochemical data, thereby enhancing sustainable soil management, climate mitigation, and cost-efficient BC-soil application through AI-driven decision-making.

5.2. Soil remediation

BCs improve HMs immobilization in soil (Das et al., 2020), and their various functional groups (e.g., –CH, –NH, –COO, and –OH) interact with pollutants, facilitating soil remediation. BCs with high regeneration ability and loading capacity are useful for pollutant immobilization in soil (Sadeh et al., 2017). For example, chicken manure-derived BC immobilized Cd²⁺ through adsorption and precipitation (Das and Avasthe, 2020) and soybean stover and pine needle BCs removed 88% Pb²⁺ and 87% Cu²⁺ from contaminated soils (Ahmad et al., 2016). Non-labile carbon fraction in soil was enhanced using BCs produced at 700 °C (Jegan et al., 2020). Different manures decreased Cr⁶⁺ in farmland soils with poultry, sheep, and cow manures having different Cr⁶⁺ (100 mg kg⁻¹) efficiencies in acidic (61.5, 57.8, and 66.6 mg kg⁻¹) and alkaline (73.9, 68.1, and 58.7 mg kg⁻¹, respectively) soils (Mandal et al., 2017).

BCs accelerate adsorption and degradation of organic compounds, such as antibiotics (Yue et al., 2019), polycyclic aromatic hydrocarbons (PAHs) (Li et al., 2019), and phthalate esters (PAEs) (Chen et al., 2021), from contaminated soils. Cow manure- and four plant-derived BCs were added to soil contaminated with tetracycline (TTC), oxytetracycline (OTC), and chlortetracycline (CTC), and incubated for 60 days in the dark (Yue et al., 2019). Cow manure-derived BC significantly enhanced antibiotic removal rates by 10.86% (TTCs), 10.29% (OTCs), and 10.12% (CTCs) compared to control. This improvement was due to enhanced microbial accessibility from higher electrical conductivity and stimulated microbial activity from BC-driven carbon and nitrogen. Sawdust- and wheat straw-derived BCs produced at 300 °C and 500 °C were applied to petroleum-contaminated soil and incubated for 84 days to assess their efficacy in removing PAHs (Kong et al., 2018). 500 °C BC significantly accelerated PAHs biodegradation, likely due to its favorable properties that enhance microbial habitat conditions. High-molecular-weight PAHs, owing to their hydrophobic aromatic rings, were more resistant to microbial degradation than low-molecular-weight PAHs (Kong et al., 2018; Li et al., 2019). BCs derived from agro-industrial waste effectively immobilized PAEs, reducing their

bioavailability while improving soil properties (Chen et al., 2021; He et al., 2016). Key factors influencing PAEs removal include BC feedstock type, pyrolysis temperature, application rate, BC pH, BC aging, and SOC content (He et al., 2016; Zhang et al., 2016). Adsorption efficiency depended on both soil characteristics (e.g., organic matter and texture) and BC properties. Notably, aging of bamboo-derived BC and SOC levels significantly affected PAEs adsorption in amended soils (Zhang et al., 2016).

Recent advances in ML/DL frameworks have significantly enhanced predictive modeling and classification for pollutant immobilization in soil remediation, enabling efficient, scalable, and sustainable data-driven strategies for complex scenarios (Guo et al., 2023). RFs were effective in predicting HMs immobilization ratios in BC-treated soils, achieving high accuracy (Testing R^2 : 0.85) and experimental validation (<20% error). Application of ML frameworks for predicting HMs immobilization in BC-amended soils showed that properties of BC were the most important, followed by experimental and soil properties. Nitrogen content (0.3–25.9%) and application rate (0.5–10%) were the key factors influencing HMs immobilization for soil remediation (Palansooriya et al., 2022). Yang et al. employed GB-DTs and SHAP-based feature attribution to model adsorption behavior of six HMs (Cd²⁺, Cr⁶⁺, Cu²⁺, Pb²⁺, Ni²⁺, and Zn²⁺) based on a large multi-source dataset. This data-driven approach enabled the development of global maps of adsorption capacity, emphasizing the importance of ML in large-scale environmental risk assessments (Table 2) (Yang et al., 2021). GB-DTs outperformed the other models in predicting HMs adsorption on soil, while SHAP clarified feature importance, leading to more accurate metal-specific models. ANNs and RFs models outperformed traditional linear models (R^2 : >0.84) in predicting HMs remediation for 930 datasets. RFs were more forgiving of missing data, and model interpretability showed the effects of metal type, BC pH, dose, and remediation duration (Sun et al., 2022). A novel feature, (H-O-2N)/C, was introduced to interpret the GBR for HMs adsorption onto BC, representing hydrogen sites linked to carbon that are non-active sites for HMs (Shen et al., 2024). Along with BC pH, this feature effectively replaced CEC in predictions and expanded variable coverage by incorporating additional cases, validated further with cases lacking CEC and specific surface area (R^2 : 0.78) and experimental results (R^2 : 0.72). Four predictive ML frameworks, namely linear regression (LR), PLSR, SVR, and RFs, were compared to assess adsorption/immobilization efficiency of BC-HMs for effective remediation of soil-plant environments (Li et al., 2024a). Moreover, a DL-integrated system was used to predict soil toxicity and reclamation decisions, which was significant for large-scale environmental assessments (Naizabayeva et al., 2024).

A hybrid framework combining reactive transport modeling and ML was developed to cost-effectively select optimal remediation strategies under dynamic constraints, successfully validated at arsenic- and PAHs-contaminated sites (Zhang et al., 2024). AutoML enabled automated prediction of optimal conditions for bioremediation of petroleum hydrocarbon-contaminated soil (Wang et al., 2024). RFs (R^2 : 0.90) achieved higher predictive accuracy for PAHs thermal desorption than ANNs and SVMs, and heating conditions (52%) were the most influential factor in model output than soil (20%) and PAHs (28%) properties on remediation efficiency (Chen et al., 2024). Composting bioremediation and BC immobilization effectively enhanced petroleum hydrocarbon degradation, outperforming bioaugmentation and phytoremediation. Bioremediation success depended primarily on hydrocarbon concentration and cultivation duration, not hydrocarbon type or soil properties, with optimal efficiency below 5000 mg kg⁻¹ (Wang et al., 2024). Six data-driven ML frameworks were developed: RFR, Gradient Boosting Regression Tree (GBRT), XGBoost, MLP, SVR, and k-NN, to unravel the complex dynamics of PAEs pollution in soil (Pan et al., 2025). An optimized XGBoost, which outperformed three other frameworks in predicting PAHs remediation efficiency by integrating key factors, offered improved accuracy to guide thermal desorption applications in soil remediation (Zhang et al., 2024d). Furthermore, an integrated

metagenomic modeling approach provided insights into the effects of BC on microplastic degradation in flooded soils (Ji, 2024). Despite these outcomes, ML/DL frameworks have significant potential to enhance BC applications in soil remediation, enabling improved pollutant removal and supporting sustainable agriculture.

Microbial communities and enzyme activities in soil are crucial factors influenced by BCs, which alter environmental characteristics and toxic element concentrations (Pan et al., 2023). The impact of BC on microorganisms is complex and depends on its physicochemical properties and environmental conditions. The type and amount of BC have varied effects: low BC levels may enhance microbial communities, whereas high levels may suppress enzyme activities (Huang et al., 2017). Compared with original biomass, BC showed a lesser effect on bacterial communities, as discovered by 16S rRNA gene pyrosequencing (Ahmad et al., 2016). Excessive BC application can disrupt C/N stoichiometry, degrade soil structure, disturb microbial balance, and impair soil function (Mukherjee and Lal, 2014). It may also trigger microbes to produce more nutrient-acquiring enzymes, intensifying competition for nutrients between plants and microbes (Tian et al., 2022). Metadata of 525 ITS and 1288 16S rRNA sequencing samples using ML revealed that BC significantly altered soil bacterial and fungal communities, enriching beneficial taxa, microbial robustness, and bacterial biosynthesis, increasing saprophytic fungal abundance and diversity for improved nutrient cycling, while reducing fungus-bacteria interactions (Lei et al., 2023).

6. ML/DL frameworks and economic feasibility perspectives of BC-soil applications

Economic feasibility of BC considers costs such as feedstock collection, processing, post-treatment, transport, and revenue from their value, GHG emissions reduction, and NO_x mitigation (Dutta and Raghavan, 2014). Commercial viability and sustainability of BC systems rely on extensive economic and environmental evaluations (Zeymer et al., 2017). Conventional CBA and LCA for BC assessment are data-intensive, time-consuming, and computationally burdensome; integrating ML/DL enhances prediction, optimization, and scenario analysis, improving decision-making and scalability (Khandelwal et al., 2024; Olafasakin et al., 2021).

Studies have shown that BC production costs of two manufacturing processes of 2211 and 11056 t/year ranged from €150 to €230/t (US \$170 to US\$257/t) for municipal green waste in Germany, with raw material costs accounting for 22–31% of total operating costs (Zeymer et al., 2017). For compost and grapefruit HTC, BC cost was US\$226/t at 7000 t/year production from biomass of 14000 t/year (Lucian and Fiori, 2017). Commercial BC cost varied from US\$750/t for powder to US\$2000/t for granular (Lawrinenko et al., 2017), but lignocellulosic biomass-derived BC is economical due to its low feedstock costs (US \$11.3–US\$24.8/t) (Daystar et al., 2014). Reinforcement learning algorithms apply pyrolysis parameters by adjusting them to balance economic viability with beneficial environmental effects, such as minimizing GHG emissions (Ascher, 2024). In addition, DL frameworks increase precision of carbon LCA by quantifying sequestration dynamics under diverse soil characteristics (Quijano et al., 2024). These studies highlight ML/DL implementation bridges data science and BC science to promote sustainable resource management, turning BC into a cornerstone of climate-smart agriculture and industry.

6.1. ML/DL frameworks for CBA

BC systems use CBA to evaluate capital and operational costs, revenue, and external factors to guide investment and policy decisions in any project, with their commercial viability depending on economic feasibility (Leiva Vilaplana et al., 2023). Due to extensive experimental evaluations, traditional techno-economic assessments (TEA) are time-consuming and cost-intensive (Zimmermann et al., 2018). Increasing

popularity of ML/DL frameworks has been leveraged to improve CBA by finding optimal process parameters, accurately forecasting production costs, and predicting market trends with high accuracy (Buja, 2023). A study in Brazil examined effects of BC on forage grass productivity (*Brachiaria* spp.), wherein BC increased yield by 27% over 2 years and improved soil pH, CEC, and macronutrient levels (Latawiec et al., 2019). Each hectare with BC saved 91 t CO₂-eq via land sparing and sequestered 13 t in soil, valued at US\$455 in carbon credits. However, high labor costs have made BC production expensive for smallholders. CBA and LCA evaluates BC's potential to replace fertilizers and sequester carbon but must account for evolving nutrient speciation, such as non-available nitrogen forms and NO_x emissions, to accurately assess nutrient availability and long-term carbon storage (Roberts et al., 2010). In Nepal, CBA of BC application in silty loam soil over 3 years (6 cropping seasons) to measure maize (93%) and mustard (134%) yields and soil properties found 15 t ha⁻¹ (dose range: 0–40 t ha⁻¹) as optimal dose across carbon price scenarios, boosting gross margin by 21% (without carbon credits) to 53% (with US\$42/t CO₂) (Pandit et al., 2018). Yield increases in 2nd and 3rd years correlated strongly with improved soil pH, available phosphorus, potassium, SOC, CEC, and base saturation. CH₄ emissions during BC production accounted for 30% of costs, while health costs were negligible.

Key advantage of using ML/DL for CBA is the ability to combine different financial metrics, including capital expenses, operation expenses, net present value, internal rate of return, and payback period (Leiva Vilaplana et al., 2023). Training ML algorithms on historical data and real-time market information can reduce uncertainty in BC production economics and generate predictive insights into BC production economics (Akinpelu et al., 2023). For example, RFs have been used to predict BC yield and economic feasibility under different pyrolysis conditions with higher estimation accuracy than traditional economic models (Ma et al., 2023). In addition, to improve prediction accuracy of costs, the relationship between BC production parameters and market conditions is modeled using nonlinear DL frameworks (e.g., ANNs and RNNs) (Akinpelu et al., 2023). However, in recent BC sales and GHG markets with negligible emission prices, ANN-based models accurately predicted revenue generation from BC sales and GHG offsets across various climatic scenarios, facilitating a holistic economic assessment. From a market perspective, ML/DL frameworks can evaluate consumer demand, regulatory impacts, and supply chain efficiencies, which are important for scaling BC-based enterprises (De Bakker et al., 2005). Integration of ML-driven TEA with LCA further improves BC's sustainability evaluation and makes it more suitable for policy formulation and industry adoption (Chitkeshwar, 2024; Nguyen et al., 2024b). Although ML can provide many benefits for CBA in BC production, it also faces challenges related to data scarcity, model interpretability, and computational resource burden (You and Wang, 2019). By developing open-access BC economic databases and integrating explainable AI frameworks to address these challenges, it is possible to develop more reliable and transparent ML-driven economic assessments (Mafat et al., 2024; Popescu et al., 2024).

6.2. ML/DL frameworks for LCA

LCA is a critical tool for environmental impact assessment of BC production, carbon balance, and its application along its value chain (Dutta and Raghavan, 2014). In Vietnam, LCA evaluated climate impacts of rice production via BC application using available straw and husk. CH₄ emissions from soil mainly contributed to carbon footprint due to BC application rates. In the 1st year, carbon footprint was reduced by 26% for spring rice and 14% for summer rice, and these further reduced to 49% and 38%, respectively, after 8th year (Mohammadi et al., 2016). Implementing negative emissions technologies across LCA can reduce GHG emissions, with effective varying techniques, ranging from 603 kg CO₂-eq./t for biomass building supplies to 1173 kg CO₂-eq./t for BC soil amendment (Jeswani et al., 2022). LCA was conducted to

evaluate environmental impacts of BC-soil systems, using 1 ton of BC produced from agriculture/forestry residues pyrolyzed at 300, 400, and 500 °C (Muñoz et al., 2017). Using SimaPro, climate change was the most significant impact category, with up to 2.74 t CO₂-eq. emissions avoided per ton of BC applied to soil.

ML/DL has expanded beyond traditional yield and kinetics predictions to novel domains like LCA of pyrolysis, leveraging its ability to model nonlinear relationships and adapt to diverse feedstocks and process conditions, thereby reducing uncertainty (Cheng et al., 2020; Khandelwal et al., 2024). For example, Cheng et al. developed RFs to predict BC yield and properties for further applications in LCA. Model showed satisfactory results consistent at lab-scale (R^2 : 0.78–0.87) and pilot-scale (R^2 : 0.45 to 0.66). Lignocellulosic biomass results yielded positive values, with a negative global warming potential (from –1050 to –200 kg CO₂-eq/t) and favorable energy returns on investment (1.9–4.3), primarily due to CO₂ sequestration by BC (Cheng et al., 2020). Olafasakin et al. designed a kriging-based reduced-order ML model with 400 experimental data points from 314 feedstocks. Model showed good yield predictive results with R^2 of 0.99. Model was applied adequately on simulations at commercial scale of 2000 t/day in Aspen Plus (RMSE: 0.1633) (Olafasakin et al., 2021). Fuel prices for minimum production scenario were US\$2.62 to US\$5.43 per gallon, and corresponding life-cycle GHG emissions were between –13.62 and 145 kg CO₂ M^{–1} J^{–1} (Olafasakin et al., 2021).

ML and LCA framework used five models to predict BC yield, carbon, and nitrogen content while evaluating pyrolysis parameter effects on environmental footprints, such as global warming potential, cumulative energy demand, and eutrophication, with sensitivity analysis quantifying their influence on global warming potential (Li et al., 2025). In contrast, carbon savings ranged up to –1355 kg CO₂-eq/t feedstock under fertilizer substitution scenarios. RFs and meta-analyses have studied the long-term stability of BC in soil systems to improve climate impact assessments (Dang et al., 2022). ML and LCA can also determine energy-efficient configurations based on different pyrolysis techniques that minimize environmental impact while maximizing BC yields (Farghali and Osman, 2024). Life cycle optimization models can be developed to assess the trade-offs between energy use, emissions, and BC quality using ML algorithms to guide process design improvements (Jie, 2022). GBR and SVR accurately trained models to estimate GHG emissions, energy consumption, and land use changes by training models on historical LCA datasets (Wang et al., 2024). Likewise, DL frameworks address complex issues by filling data gaps and reducing uncertainties in LCA emission estimates. A DL framework, explainable AI internet of things (IoT), has been employed to model interactions among BC properties, soil conditions, and microbial activity to understand role of BC in mitigating N₂O and CH₄ emissions (An, 2024).

Standardization gaps, data heterogeneity, and computational complexity hinder widespread adoption; interdisciplinary collaboration among data scientists, environmental engineers, and policymakers is essential to build robust ML/DL-enhanced LCA frameworks, improving reliability, CBA, and sustainability in BC-soil applications.

7. Challenges, limitations, and their potential solutions

Along with complex BC and soil properties, data scarcity and poor data quality, limited interpretability and transparency of frameworks, high computational requirements and scalability, model bias and generalization, difficulties in feature selection and dimensionality, temporal and spatial variability, and limited field and practical applications are key challenges that prevent the use of ML/DL frameworks in BC-soil applications (Table 3). Inconsistencies in BC feedstocks, preparation parameters, carbon content, application rates, and aging can be optimized by producing and storing BC under optimal conditions to enhance efficiency in soil applications, while maintaining optimal pH, temperature, and moisture levels based on prior soil data to account for variations in texture, mineralogy, and C/N ratios of agricultural and

Table 3

Challenges, limitations, and potential solutions for implementing ML/DL frameworks in BC-soil applications.

Challenges	Implications/limitations	Potential solutions	Sources
BC properties	Significant variability in feedstock type, preparation parameters, carbon content, particle size, and aging affects reproducibility and model accuracy	Optimize pyrolysis conditions (e.g., 400–500 °C for 1.5–2 h) and standardize feedstock characterization	(Tisserant and Cherubini, 2019)
Soil properties	Heterogeneity in pH, texture, mineralogy, moisture, temperature, and C/N ratio complicates model generalization	Incorporate site-specific soil data and use normalization/feature scaling; model calibration with historical datasets	(Kocsis et al., 2022; Tisserant and Cherubini, 2019)
Data scarcity and quality	Limited high-quality datasets and imbalanced data lead to overfitting, reduced generalizability, and biased predictions	Apply synthetic data generation (e.g., GANs), transfer learning, data augmentation, and integrate real-time sensor data	(Waqas et al., 2025)
Interpretability and transparency	ML/DL frameworks often function as black boxes, limiting stakeholder trust and adoption in practice	Utilize SHAP, LIME, attention mechanisms, physics-informed ML, and prioritize interpretable models	(Barragan-Montero et al., 2022)
Computational requirements and scalability	High computational costs and resource-intensive training hinder adoption in small-scale research institutions	Employ cloud computing platforms (AWS, Google Cloud), knowledge distillation, quantization, and pre-trained model adaptation	(Duan et al., 2022)
Bias and generalization	Models trained on biased or limited datasets may fail to generalize across soil types, BC formulations, or geographic regions	Collect diverse datasets, implement k-fold cross-validation, domain adaptation, and ensemble modeling	(Waqas et al., 2024)
Feature selection and dimensionality	High-dimensional input features (soil properties, BC characteristics, and environmental factors) increase model complexity and risk of overfitting	Apply feature selection, dimensionality reduction (PCA and autoencoders), and sensitivity analysis	(Jia et al., 2023; Shafizadeh et al., 2023; M. Zhang et al., 2024)
Temporal and spatial variability	Dynamic environmental conditions over time and space reduce model predictive stability and robustness	Incorporate spatiotemporal datasets, and use RNNs or LSTM	(Luo et al., 2023; L. Wang et al., 2023)
Limited field and practical applications	Most ML frameworks remain theoretical, with limited translation to field-scale BC applications	Develop digital twin frameworks, decision support systems, and pilot-scale experiments for real-time monitoring	(El-Azazy et al., 2024; Waqas et al., 2024)

farm soils (Kocsis et al., 2022; Tisserant and Cherubini, 2019). Data scarcity and the need for high-quality standardized datasets are major concerns, as data heterogeneity compromises model generalizability, and overfitting is a significant issue. GANs-based synthetic data generation, transfer learning, and real-time data acquisition via sensors have been proposed to solve this problem (Waqas et al., 2025). It impedes interpretability and trust, as they are “black box.” SHAP, LIME, attention mechanisms, and hybrid frameworks that combine physics models also increase transparency. Integrated teams, AutoML tools, and educational outreach are required to eliminate these limitations and increase interdisciplinary collaborations (El-Azazy et al., 2024; Waqas et al., 2024). In addition, computational and scalability constraints prevent access to smaller institutions. To address this, cost-effective strategies include model compression, transfer learning, and cloud computing (Duan et al., 2022). High-dimensional input features, such as soil/BC properties and environmental factors, increase model complexity and overfitting risk. This can be mitigated through feature selection, dimensionality reduction (e.g., PCA and autoencoders), and sensitivity analysis (Shafizadeh et al., 2023; M. Zhang et al., 2024). Temporal and spatial variability in environmental conditions undermine framework stability and robustness, which can be addressed by incorporating spatiotemporal datasets and recurrent architectures (e.g., RNNs and LSTM) (Luo et al., 2023; L. Wang et al., 2023). Dataset bias, lack of standardization, and limited field integration hinder model generalizability, reproducibility, and real-world applicability (Waqas et al., 2025), necessitating strategic collaborations and scalable infrastructures for pilot- and large-scale trials. Overall, resolving above-mentioned challenges requires multiple advanced frameworks, such as explainable AI frameworks, GANs, interpretable models, AutoML, and standardized protocols.

8. Conclusion

Integration of ML/DL frameworks for production, characterization, and soil application of BC represents a transformative paradigm in sustainable agricultural research. Studies have demonstrated that ML/DL frameworks, including RFs, SVMs, ANNs, DNNs, and CNNs, can effectively predict BC yield, surface area, adsorption capacities, and environmental remediation performance using diverse independent variables. ML/DL frameworks have enhanced data processing efficiency, automated real-time analysis of spectroscopic and imaging datasets, and enabled precise modeling of the impact of BC on nutrient availability, microbial activity, HMs immobilization, and GHG mitigation. Consequently, BC utilization in climate-smart agriculture has increased recently.

Despite these advancements, there are still significant limitations. Heterogeneity of data, limited model generalization, lack of standardized evaluation procedures, and underutilization of unsupervised and reinforcement learning techniques limit wider adoption. To overcome these limitations, synthetic data generation (e.g., GANs) and explainable AI (e.g., SHAP and LIME) should be adopted, and interdisciplinary collaboration should be encouraged in future studies. Moreover, it is recommended to incorporate IoT sensors, remote sensing, and cloud computing platforms into agricultural activities to facilitate the real-time implementation and testing of predictive models.

ML/DL frameworks offer an efficient, scalable, and economical approach to optimize BC technologies to support data-driven decision-making for soil management, waste valorization and carbon sequestration. The real-world application of these frameworks can accelerate market adoption of BC, support regulatory compliance, and advance sustainable development goals.

CRediT authorship contribution statement

Muhammad Waqas: Writing – original draft, Methodology, Data curation, Conceptualization. **Mohsin Nawaz:** Writing – review & editing. **Rongli Yang:** Software, Data curation. **Waseem Tariq:** Writing –

review & editing. **Muhammad Nauman Tahir:** Software, Investigation. **Shakeel Ahmad:** Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsoil.2026.106908>.

Data availability

Data will be made available on request.

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